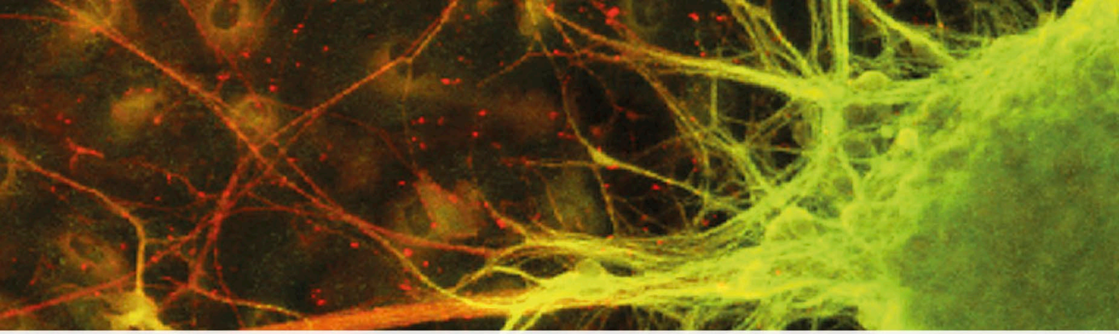

a crowded place. Tissues and organs press against one another. Body parts shift continually as we move, breathe, pump blood, and digest food. For example, swallowed food does not simply fall down the gullet into the stomach; it is forced down by waves of muscular contraction.

PHYSIOLOGY

For a rounded understanding of the body, we need to see human anatomy in combination with physiology – the study of how the body functions. Physiology focuses on the dynamic chemical minutiae at atomic, ionic, and molecular levels. It investigates the workings of such processes as enzyme action, hormone stimulation, DNA synthesis, and how the body stores and uses energy from food. As researchers look closer, and unravel more biochemical pathways, more physiological secrets are unlocked. Much of this research work is aimed at preventing or treating disease.

HEALTH AND ILLNESS

Medical science amasses mountains of evidence every year for the best ways to stay healthy. At present, an individual's genetic inheritance, which is a matter of chance, is the given starting point for maintaining health and well-being. In recent years, treatments such as pre-implantation genetic diagnosis (PIGN) – carried out as part of assisted reproductive techniques, such as in vitro fertilization (IVF) – and gene therapy are able to remove or negate some of these chance elements. Many aspects of upbringing have a major impact on health, including factors such as diet and whether it is too rich or too poor. The body can also be affected by many different types of disorders, such as infection by a virus or bacteria, injury, inherited faulty genes, or exposure to toxins in the environment.

COMMUNICATION NETWORK

This microscopic image of nerve cells (neurons) shows the fibres that connect the cell bodies. Neurons transmit electrical signals around the body; each one links with hundreds of others, forming a dense web.